K-Field Materials for Coherently Controlled Angular Momentum Density

Candidate molecules, crystals and spin systems for electromagnetically forced microscopic circulation

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This paper defines a materials-selection framework for generating coherent, reversible microscopic angular momentum with electromagnetic fields. It integrates established molecular, phonon and magnon control methods with the K-Field Base Cell geometry used to specify orientation, phase planes and handedness-sensitive measurements.

Contents

Abstract

The objective is to identify matter that couples strongly and controllably to electromagnetic fields while supporting a coherent microscopic state with nonzero angular momentum. The useful source is not simply a material that absorbs strongly. It must permit a well-defined circular or helical motion, a reproducible phase, a reversible handedness, a high participating number density and a measurement protocol that separates angular-momentum-odd effects from heating, radiation pressure, dielectric stress and ordinary magnetic coupling. The analysis finds two complementary primary candidates. Cubic $SrTiO_3$ driven by quadrature terahertz fields is the strongest direct realization of a solid-state ionic orbit. Yttrium iron garnet is the strongest practical platform for dense coherent spin angular momentum. Methyl fluoride provides a comparatively simple, strongly polar free molecular rotor; optical-centrifuged O_2 or N_2 provides much higher rotational frequency; $LiNbO_3$ provides electrically reversible chiral lattice modes; and ultracold RbCs provides exceptional coherence at low density. The report derives the control equations, maps the source axis into the complete 27-node K-Field Base Cell and seven-sheet plane family, documents measured operating data, ranks the materials by experimental role, and defines a staged test architecture.

Principal result

For the first solid-state orbital experiment, use a [001]-oriented cubic $SrTiO_3$ specimen with two orthogonal in-plane terahertz field components at approximately 3 THz and a relative phase of +90 or -90 degrees. For the first spin-selective comparison, use YIG under bias-field-tuned ferromagnetic resonance with two orthogonal microwave magnetic-field components in quadrature. For the first free molecular rotor comparison, use low-pressure CH_3F near its approximately 51.071 GHz rigid-rotor J=0 to 1 frequency. The primary observable is the difference between opposite drive handedness at equal deposited energy.

1. Experimental objective and state variable

The experiment seeks a source in which microscopic matter executes coherent circular motion under electromagnetic control. The orbit may be a real-space ionic trajectory in a solid, rotation of a molecular frame, collective precession of electronic spins, or a coherent superposition whose laboratory-frame electric dipole rotates around a quantization axis. These states differ physically, but each carries a signed axial angular momentum that can be reversed without changing the nominal drive frequency.

The central discrimination variable is handedness. Two orthogonal coordinates with equal amplitudes and a quarter-cycle phase offset form a circle:

```
q_x(t) = q0 cos(omega t) q_y(t) = s q0 sin(omega t) s = +1 or -1
```

For an effective moving mass M_eff, the signed angular momentum normal to the orbit plane is

```
L_z = M_eff [q_x dq_y/dt - q_y dq_x/dt] = s M_eff omega q0^2.
```

The sign s changes when the relative phase changes from +90 degrees to -90 degrees. Frequency, amplitude and time-averaged quadratic drive power can remain unchanged. A response that reverses with s is therefore distinguishable from most thermal and scalar electromagnetic backgrounds.

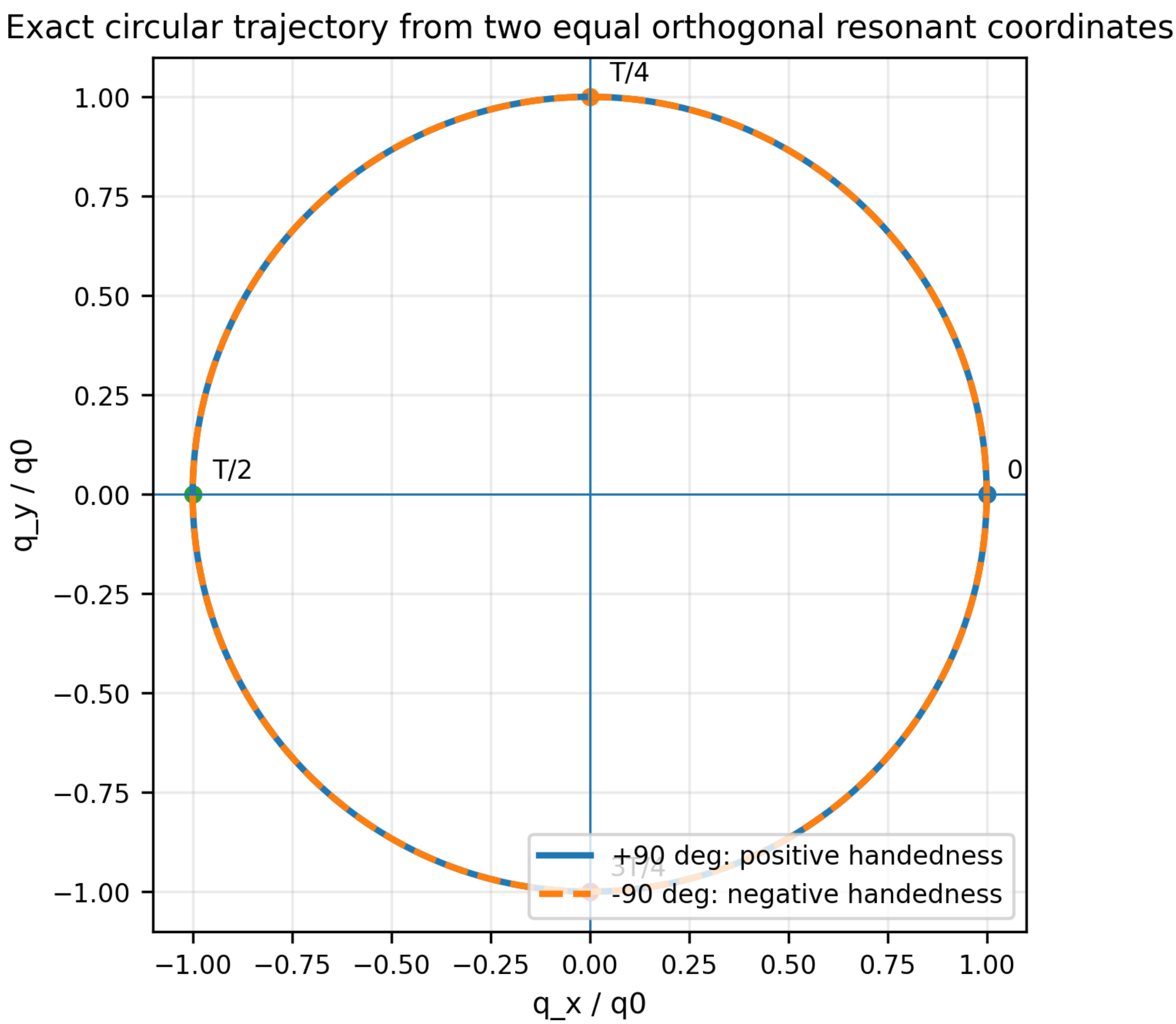

Figure 1. Exact phase-space orbit generated by two equal orthogonal coordinates. Reversing the quadrature phase reverses angular momentum while preserving the orbit radius and angular frequency.

1.1 Angular-momentum density rather than absorption

A useful material maximizes the coherent signed angular momentum per active volume. For N identical coherently phased microscopic units in volume V, a classical screening form is

```
J_coh = (N/V) M_eff omega q0^2 C_phi C_pop,
```

where C_phi is the phase-coherence factor and C_pop is the fraction of units participating in the driven mode. The expression is a control metric, not a universal material constant. It shows why a dense solid with picometer ionic displacement can compete with a gas whose individual molecules rotate over a much larger radius: the solid contains vastly more synchronized units per volume. It also shows why long coherence alone is insufficient if the participating density is very small.

1.2 Required experimental reversals

- **Handedness reversal:** +90 degrees versus -90 degrees at fixed frequency and field amplitude.

- **Linear-polarization control:** 0 or 180 degree phase difference, which preserves oscillation but removes the circular orbit.
- **Off-resonance control:** detune while preserving pulse energy and instrumentation state.
- **Axis reversal:** rotate the specimen or bias axis by 180 degrees relative to the K-Field reference orientation.
- **Material control:** use a matched specimen that absorbs comparable energy but cannot support the selected circular mode.

2. K-Field Base Cell and orientation planes

The microscopic source must be assigned a reproducible direction in the K-Field geometry. The student-facing geometric object is the K-Field Base Cell. The complete first-level centered address set contains 27 nodes indexed by L=(i,j,k), with i,j,k in {-1,0,1}. Physical node positions are generated by

```
x_L = (1/2) B_nm L,
```

where the columns of B_nm are the ordered edge vectors a, b and c in nanometers:

```
B_nm = [[0.334179679060, 0.061973646450, 0.131477878069], [0, 0.583126915466,
0.494308507783], [0, 0, 0.605097741520]] nm.
```

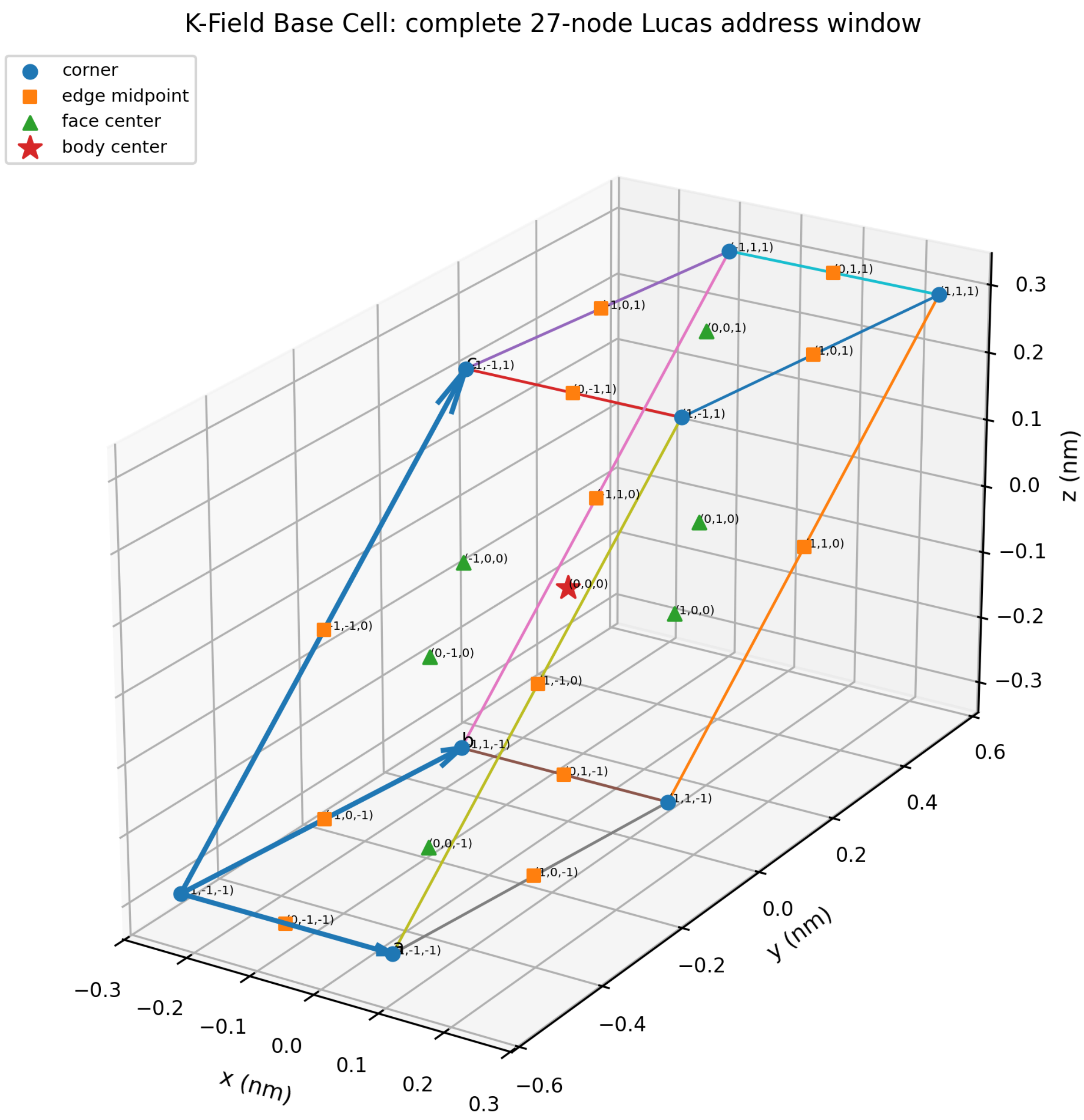

Figure 2. Dimensionally scaled K-Field Base Cell and complete 27-node centered Lucas address window. The eight corner nodes, twelve edge midpoints, six face centers and body center are generated directly from x_L=(1/2)B_nm L. The appendix gives the full coordinate ledger.

2.1 Metric properties

Quantity	Value	Meaning
|a|	0.334179679060 nm	first base-cell edge
|b|	0.586410890413 nm	second base-cell edge
|c|	0.792319765045 nm	third base-cell edge
angle(a,b)	83.933483012 deg	included edge angle
angle(a,c)	80.448130018 deg	included edge angle
angle(b,c)	50.363231814 deg	included edge angle
A_ab	0.194869165462 nm^2	area of face spanned by a and b
A_ac	0.261106235792 nm^2	area of face spanned by a and c
A_bc	0.357809539131 nm^2	area of face spanned by b and c
Volume	0.117914891913 nm^3	absolute determinant of B_nm

2.2 Seven-sheet family

The sheet index is lambda=i+j+k. The reciprocal vector normal to the sheet family is

```
g111 = B_nm^(-T) (1,1,1) = (2.992402179608, 1.396865594231, -0.138683770577) nm^-1.
```

The plane equation is g111 dot x = lambda/2. The reciprocal magnitude is 3.305289318818 nm^-1, the full reciprocal wavelength is 0.302545376075 nm, and adjacent Lucas sheets are separated by 0.151272688038 nm. The seven node populations are 1, 3, 6, 7, 6, 3 and 1 for lambda=-3 through +3. The alternating polarity is chi=(-1)^lambda.

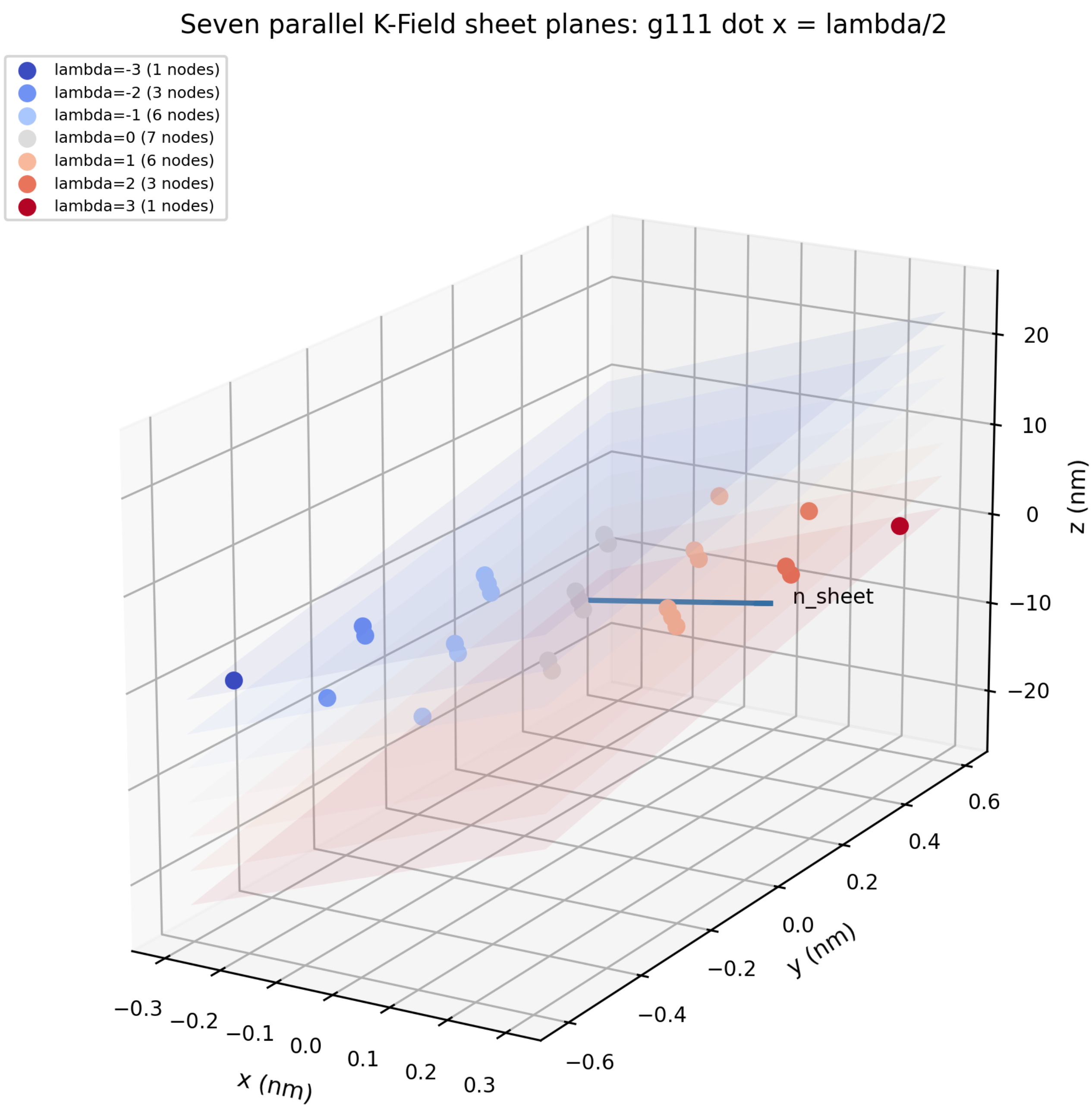

Figure 3. The seven K-Field sheet planes and all 27 nodes. The normalized sheet normal defines the preferred axis used to compare microscopic angular momentum vectors across candidate materials.

For a specimen with orbit normal n_orbit, the simplest orientation coordinate is cos(theta)=n_orbit dot n_sheet. A full orientation sweep should record the response against both n_orbit and -n_orbit, because the controlled angular momentum is time-odd and axial.

3. Materials-screening framework

The ranking uses five experimentally distinct requirements. First, the field must couple strongly enough to prepare the state without destructive heating. Second, the state must carry signed angular momentum rather than only linear displacement. Third, a macroscopic number of microscopic units must share a known phase or handedness. Fourth, the state must persist long enough for the measurement bandwidth. Fifth, the source must permit reversals that leave common-mode electromagnetic backgrounds unchanged.

Priority	Candidate	Microscopic angular momentum	Principal drive	Best role	Limiting factor
1	SrTiO3	Circular ionic soft-phonon motion	Circular THz electric field near 3 THz	Primary solid-state mass-orbit test	Picosecond duration; very strong fields
2	YIG	Collective electronic spin precession	Bias-tuned circular microwave magnetic field	Primary spin-selective test	Magnetic backgrounds require careful subtraction
3	CH3F gas	Whole-molecule rotational angular momentum	Circular microwave near 51.071 GHz initial transition	Ordinary polar molecular rotor	Collisions and lower number density
4	O2 or N2	Unidirectional molecular superrotation	Optical centrifuge; 792 nm carrier; 0-10 THz rotating polarization	Highest ordinary-gas rotational frequency	High optical peak power and pulsed operation
5	LiNbO3	Chiral lattice phonon angular momentum	THz/mid-IR excitation or ferroic domain control	Polar-crystal comparison	Mode and momentum selection are geometry dependent
6	Ultracold RbCs	Rotational-state superposition and rotating lab-frame dipole	Microwave state control in magic optical trap	Longest molecular rotational coherence	Very low active density and complex apparatus

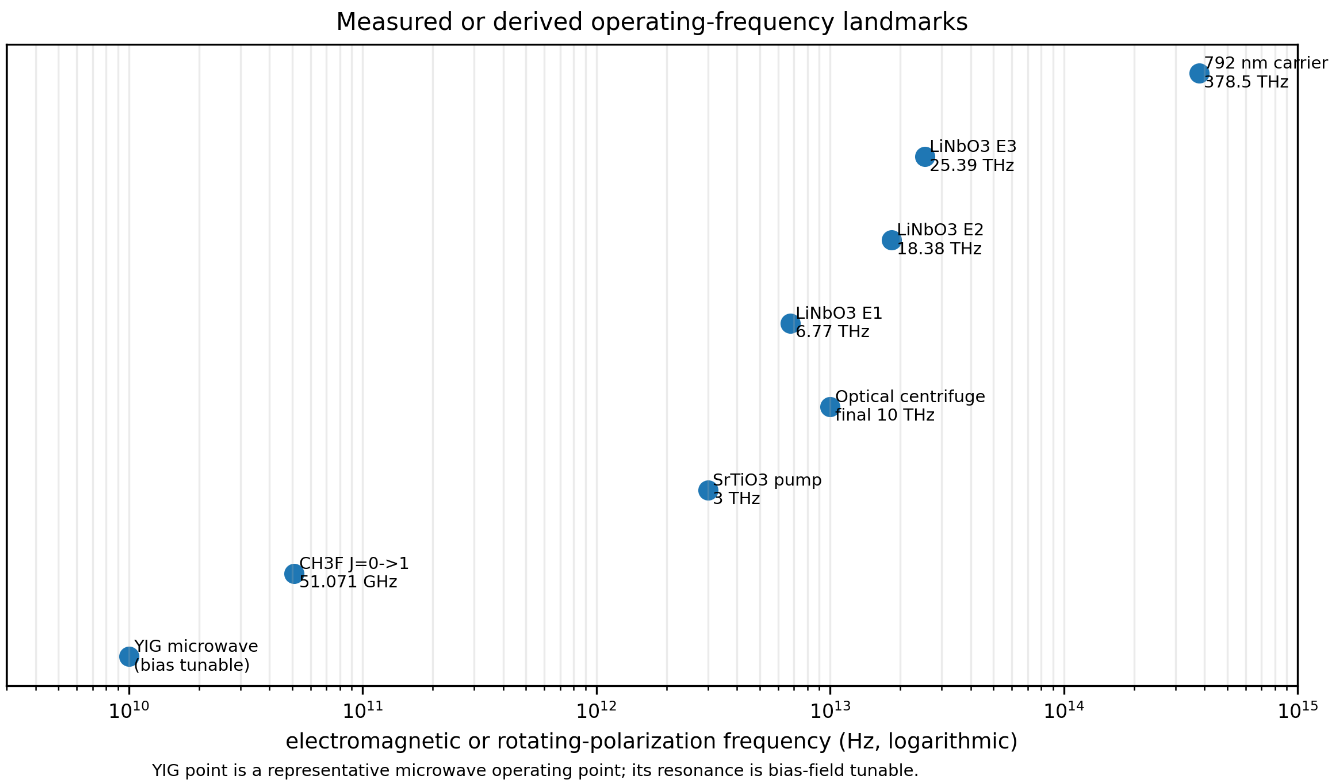

Figure 4. Frequency landmarks from the candidate data. The useful microscopic rotation frequency is not always the electromagnetic carrier frequency: the optical centrifuge uses a visible carrier while the polarization axis rotates in the terahertz range.

The ranking is role-specific. $SrTiO_3$ is first for literal ionic circulation in a dense solid. YIG is first for controllable collective spin. CH_3F is first among practical ordinary polar gases. RbCs is first for coherence time but not for angular-momentum density.

4. SrTiO3: primary solid-state ionic-orbit candidate

At room temperature SrTiO3 is cubic, paraelectric and diamagnetic. Its infrared-active soft polar phonon has three degenerate Cartesian eigenvectors, allowing two orthogonal components to be driven with equal amplitude and a quarter-cycle phase difference. The resulting ionic displacements trace closed trajectories. This is the direct solid-state realization of the orthogonal-channel concept.

The 2024 experiment drove the mode with circularly polarized terahertz pulses centered at 3 THz, with 0.5 THz bandwidth and wavelength near 100 micrometers. The reported pump field amplitude was approximately 230 kV/cm. The source was resonant near room temperature because the soft mode is approximately 2.7 THz at 300 K. The same mode softens to approximately 0.2 THz at 5 K [1].

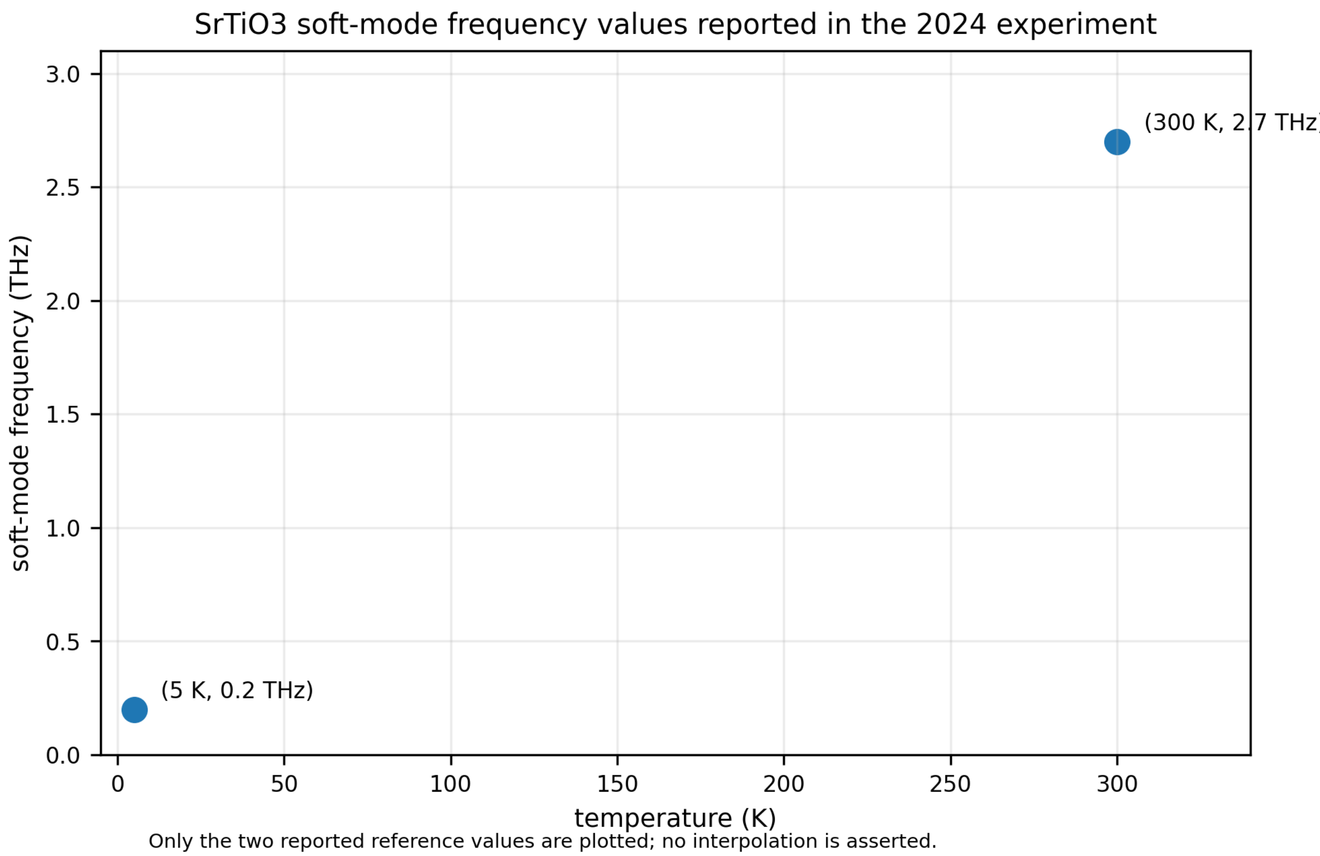

Figure 5. Two reported soft-mode frequency landmarks for SrTiO3. Temperature provides an experimental resonance-tuning coordinate.

4.1 Ionic displacement and source magnitude

The model used displacement amplitudes of approximately 1 pm for titanium and strontium and approximately 3 pm for oxygen. A coherent orbit with radius q0=3 pm and angular frequency omega=2 pi(3 THz) has a peak tangential speed of about 56.5 m/s. This is microscopic displacement with a high cycling rate and a solid-state unit-cell density.

The same study calculated a pure dynamical-multiferroic moment of order 10^-2 nuclear magnetons per unit cell, while its magneto-optical estimate was of order 10^-1 Bohr magnetons per unit cell. The authors therefore identified an approximately four-order-of-magnitude gap between pure ionic-orbit theory and the inferred experimental magnitude and discussed angular-momentum transfer to the electronic system [1]. For K-Field testing, this distinction is useful: the ionic trajectory is the prepared source, while magnetic and electronic observables independently verify that the driven mode carries handed angular momentum.

4.2 Recommended specimen geometry

Use a high-quality cubic $SrTiO_3$ crystal with the two drive-field axes aligned to two orthogonal crystallographic directions in the face. The third axis, normal to the drive plane, is the microscopic angular-momentum axis. Mount the specimen so this axis can be swept relative to n_sheet. A fixed specimen with electronically switched +90/-90 degree phase provides the fastest reversal. A second mechanical rotation stage provides an independent 180 degree axis inversion.

4.3 Measurement sequence

- Acquire +90 degree circular drive, -90 degree circular drive, linear drive and no-drive traces in an interleaved sequence.
- Normalize each pulse to measured terahertz energy and record transmitted/reflected field waveforms.
- Use a force, acceleration or torsion sensor with bandwidth below the pulse-repetition frequency and synchronous demodulation keyed to helicity.
- Record Kerr rotation or an equivalent phonon-sensitive observable to confirm that the circular mode was excited on every acquisition block.
- Sweep temperature through the resonance condition while holding orientation fixed; then sweep orientation while holding temperature and resonance fixed.

5. YIG: primary collective-spin candidate

Yttrium iron garnet supports low-loss collective magnetization precession. Under a static bias field, a microwave magnetic field perpendicular to the equilibrium magnetization excites ferromagnetic resonance. Two orthogonal microwave field components in quadrature produce circular precession. Reversing their phase order reverses the microwave helicity and the driven magnon angular momentum.

A submillimeter YIG sphere in a field-focusing microwave cavity demonstrated a normal-mode splitting of 2 GHz, cooperativity of 10^5 and an approximately 3 percent magnetic filling factor at millikelvin temperature [2]. Those numbers demonstrate exceptionally strong coherent photon-magnon coupling. They are not a fixed YIG resonance frequency: the resonance is set by the bias field, specimen geometry and mode.

5.1 Why YIG is complementary rather than interchangeable with SrTiO3

The dominant controlled coordinate in YIG is electronic spin and collective magnetization, whereas the dominant coordinate in the SrTiO3 soft mode is ionic motion. A K-Field response that tracks both systems according to total angular momentum would support a broad angular-momentum coupling. A response confined to the ionic system would favor mass circulation. A response confined to YIG would favor electronic spin or magnetic mechanisms. The pair therefore functions as a mechanism separator.

5.2 Required magnetic controls

- Reverse microwave helicity while keeping the static bias field fixed.
- Reverse the bias field and adjust microwave helicity so that the laboratory angular-momentum vector is either preserved or reversed in a controlled way.
- Use a nonmagnetic dielectric resonator or matched lossy sample to measure microwave heating and Lorentz-force backgrounds.
- Map response versus bias field across resonance, because a genuine magnon-mediated source follows the resonance envelope.

6. CH3F: primary ordinary polar molecular rotor

Methyl fluoride is a symmetric-top molecule with C3v symmetry. NIST reports rotational constants A=5.18200 cm^-1 and B=C=0.85179 cm^-1, a permanent dipole moment of 1.847 D and an electric polarizability of 2.540 cubic angstroms [3]. Its symmetry produces a simpler rotational structure than an asymmetric-top molecule such as water and its permanent dipole provides direct microwave coupling.

In the rigid-rotor approximation, the lowest J=0 to 1 spacing associated with B is 2Bc=51.072044 GHz, corresponding to a free-space wavelength of 5.869991 mm. Actual transition selection requires the symmetric-top quantum numbers and the experimental pressure, temperature and electric-field environment, but this value defines the correct first microwave design scale.

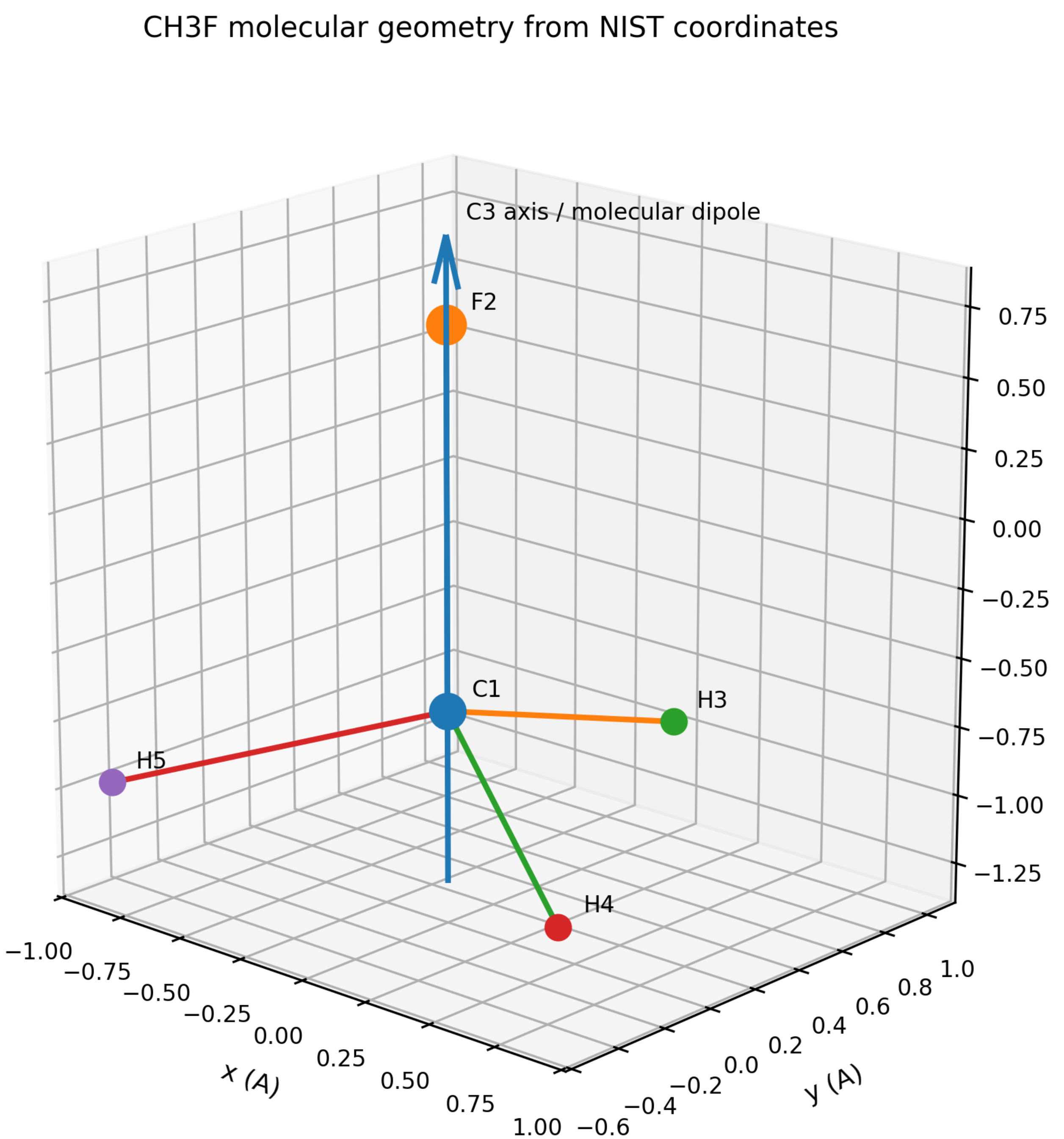

Figure 6. Methyl-fluoride geometry from NIST Cartesian data. The permanent dipole and symmetry axis lie approximately along the C-F axis. The complete coordinate set is included in the machine-readable file.

6.1 Circular microwave preparation

Two orthogonal microwave electric-field components in quadrature address Delta-M=+1 or Delta-M=-1 branches relative to the quantization axis. The preparation produces a selected projection of molecular rotational angular momentum rather than a classical ensemble in which every molecule has the same instantaneous orientation. A strong dc electric field may define the quantization axis and reduce orientation ambiguity.

6.2 Gas-cell constraints

The molecular source is limited by pressure broadening, collisions, wall interactions and the standing-wave structure of the microwave cavity. A first apparatus should use a sealed low-pressure cell inside a high-Q resonator, with pressure treated as a swept parameter. The cell should be mechanically symmetric and support reversal of microwave helicity without moving components.

7. O2 and N2 optical centrifuge: maximum ordinary-gas rotation rate

An optical centrifuge forms a linearly polarized optical field whose polarization axis accelerates in rotation. It is produced by combining oppositely chirped, oppositely circularly polarized spectral components. Molecules with anisotropic polarizability follow the rotating axis and climb the rotational ladder through Raman coupling.

The reported implementation used a carrier centered near 792 nm and accelerated the polarization rotation frequency from zero to approximately 10 THz in about 70 ps. It prepared oxygen up to rotational quantum number N=95 and nitrogen up to N=60, with directly observed coherent unidirectional rotation [4]. The visible carrier frequency is approximately 378.5 THz, but the mechanically relevant imposed rotation is the difference frequency that reaches 10 THz.

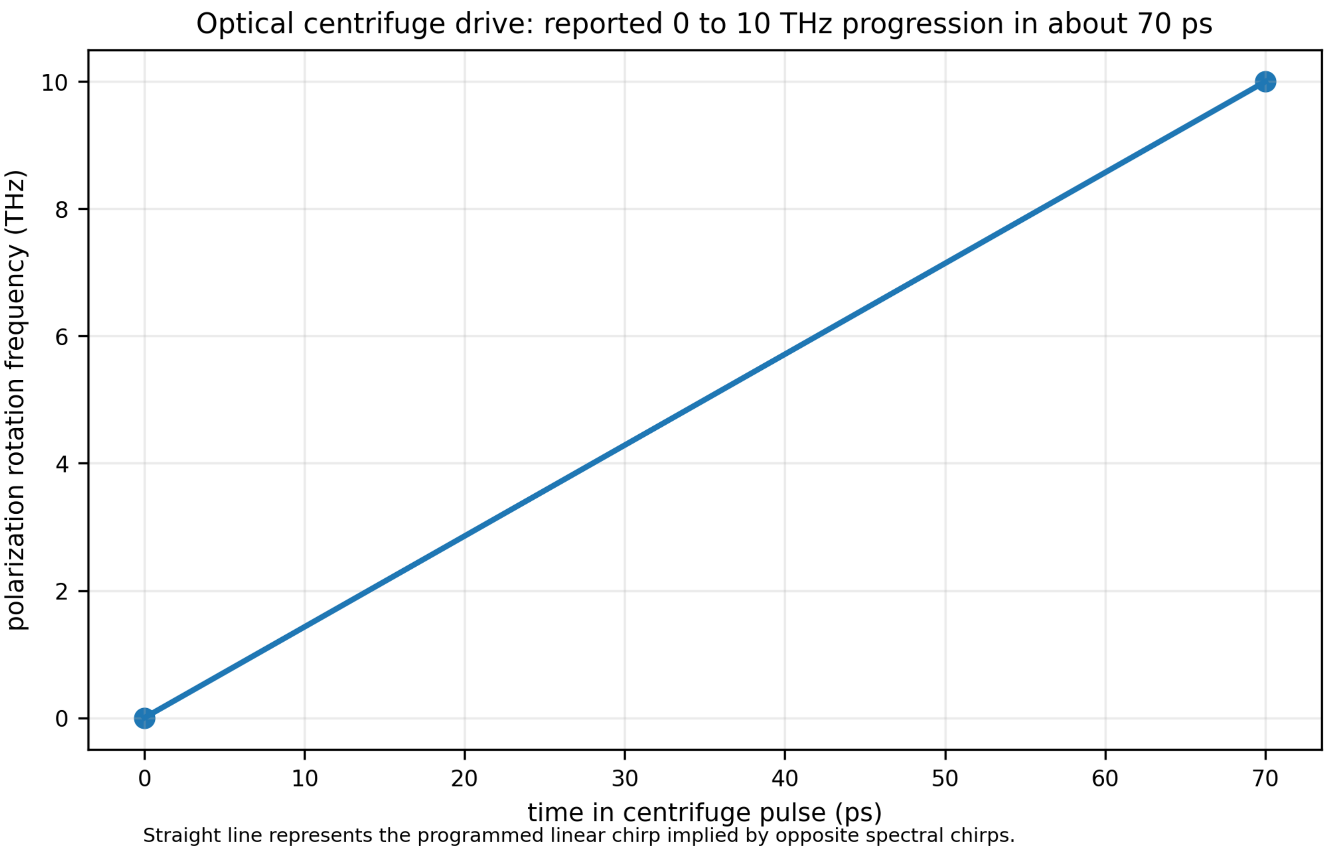

Figure 7. Programmed optical-centrifuge rotation frequency using the reported endpoints. The slope is the drive angular acceleration in frequency units.

O2 combines molecular rotation with electronic spin; N2 is a cleaner closed-shell control. A paired O2/N2 experiment can therefore separate effects that depend on rotating nuclear mass from those enhanced by intrinsic spin.

The principal disadvantage is instrumentation complexity. The source is pulsed, the optical fields are intense, and plasma, acoustic and thermal transients require aggressive common-mode subtraction. Nevertheless, it offers the highest demonstrated rotational frequency among ordinary molecular gases in this candidate set.

8. LiNbO3: polar crystal with reversible chiral phonons

LiNbO3 is a non-centrosymmetric ferroelectric in which chiral phonons have been directly observed. The reported chiral modes include representative energies near 28, 76 and 105 meV, corresponding to approximately 6.77, 18.38, 25.39 THz [5]. The phonon angular momentum reverses with handedness, and ferroelectric-domain inversion provides an additional electrical method for reversing the momentum-dependent phonon chirality.

This material is valuable because it adds a static ferroic control bit to the dynamic optical or terahertz helicity. Four states can therefore be compared: two optical helicities in each of two oppositely polarized ferroelectric domains. Effects tied to phonon chirality, domain polarity, optical helicity and deposited energy can be separated algebraically.

LiNbO3 is not the first recommended test because the chiral modes depend on wave vector and crystal symmetry, and the most direct 2026 observation used resonant inelastic X-ray scattering. It is the preferred second polar solid after SrTiO3 because its ferroelectric domains provide nonvolatile handedness control.

9. Ultracold RbCs: coherence benchmark

Ultracold 87Rb133Cs provides the longest directly demonstrated rotational coherence in this candidate set. A rotationally magic optical trap supported a Ramsey coherence time of 0.78(4) s and a spin-echo lower bound greater than 1.4 s at 95 percent confidence. The reported sample contained approximately 2,400 molecules at 1.5 microkelvin and peak density 6x10^10 cm^-3. The molecular-frame dipole was 1.23 D [6].

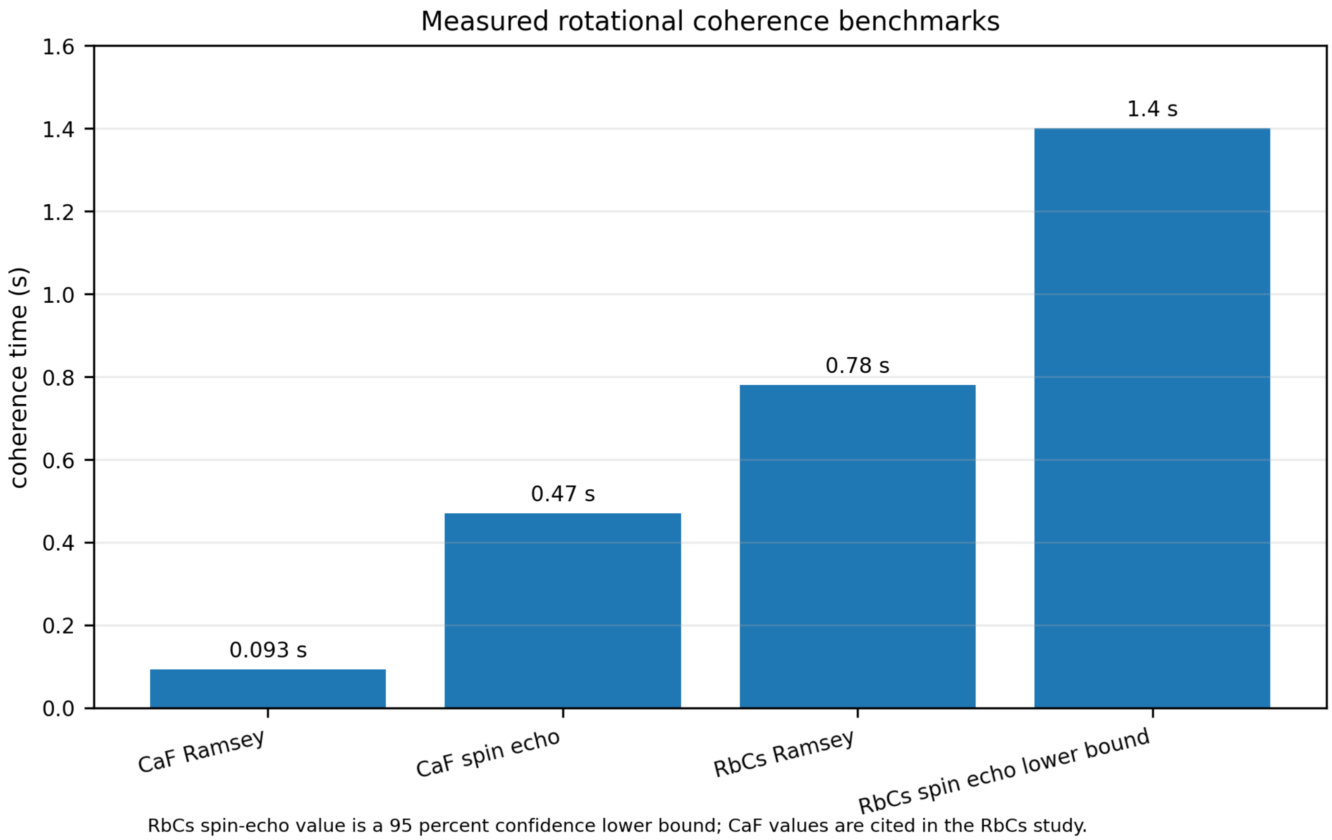

Figure 8. Rotational coherence benchmarks reported or summarized in the RbCs study. RbCs provides exceptional coherence, but its active number density is far below that of a solid.

A coherent superposition of suitable rotational states produces a laboratory-frame dipole that rotates around the quantization axis. This is a particularly clean realization of a quantum rotating source because phase, population and coherence can be measured directly by Ramsey interferometry. Its limitation for force detection is the small total mass and molecule number. RbCs is therefore best used as a coherence and selection-rule reference rather than the first high-signal apparatus.

10. Comparative interpretation of the candidate source terms

The candidate systems populate three physically distinct source classes.

10.1 Real-space mass circulation

SrTiO3, CH3F, O2, N2 and LiNbO3 involve motion of nuclei or ions in real space. The participating radius ranges from picometers in phonons to molecular bond-length scales in free rotation. In a solid the orbit radius is small but the coherent number density is high. In a gas the radius and rotational quantum number can be larger but collisions and lower density reduce the collective source.

10.2 Electronic spin circulation

YIG produces coherent collective spin precession with minimal charge transport. It is the cleanest test of whether the K-Field responds directly to spin angular momentum. O2 provides a hybrid case because the rotating molecule also has electronic spin.

10.3 Rotating quantum dipoles

RbCs and driven polar molecules need not resemble little rigid bodies with a continuously localized orientation. Their quantum states can instead produce a rotating expectation value of the dipole and a well-defined angular-momentum projection. The experimentally relevant source variables are the density matrix, state populations, phase coherence and angular-momentum expectation value.

10.4 Frequency is not the only multiplier

The elementary classical expression M_eff omega q0^2 shows that increasing frequency helps only when displacement and coherent population remain substantial. Near resonance q0 depends on oscillator strength, damping and field amplitude. At very high drive frequency, ionization, bond distortion or mode leakage can reduce coherence. The correct optimization is therefore a measured sweep of angular-momentum-sensitive response versus frequency, amplitude, coherence and orientation, not a frequency-only maximization.

11. Recommended staged experimental program

Stage 1 establishes a dense solid-state ionic source. Stage 2 separates ionic and spin coupling. Stage 3 transfers the control law to a free molecular rotor. Stage 4 explores the high-frequency and long-coherence limits.

Stage	Material and source	Primary reversal	Primary readout	Purpose
1A	$SrTiO_3$, 3 THz circular soft-phonon drive	THz helicity +90/-90 degrees	synchronous force/acceleration plus Kerr verification	detect handed ionic-orbit response
1B	$SrTiO_3$, linear and off-resonance controls	quadrature removed; frequency detuned	same sensors and pulse energy normalization	measure thermal and scalar EM backgrounds
2	YIG ferromagnetic resonance	microwave helicity and bias-field reversal	force/acceleration plus microwave S-parameters	separate spin from ionic mass circulation
3	Low-pressure CH_3F microwave resonator	Delta-M branch / microwave helicity	force/acceleration plus absorption and cavity phase	test free molecular rotation
4A	O_2 and N_2 optical centrifuge	centrifuge handedness	pulsed force/impulse and Raman rotational-state readout	extend rotation frequency toward 10 THz
4B	$LiNbO_3$ chiral phonons	optical helicity and ferroelectric domain	force/acceleration plus phonon-sensitive optical readout	test nonvolatile chirality control
4C	Ultracold RbCs	microwave phase and state superposition	precision state interferometry and local force sensor	coherence-limited reference

11.1 Helicity-demodulated estimator

For paired acquisitions with equal field magnitude and opposite handedness, define

```
S_odd = [S(+)-S(-)]/2 S_even = [S(+)+S(-)]/2.
```

S_odd isolates responses that reverse with angular momentum. S_even records heating, scalar electrostriction, average radiation pressure and other effects that do not reverse. Linear-polarization and off-resonance measurements are then used to decompose residual terms in both channels.

11.2 Orientation sweep

At each material resonance, sweep the angular-momentum axis over the unit sphere or over a mechanically accessible great-circle set. Record n_orbit in the same coordinate system as the K-Field Base Cell sheet normal. The minimum complete sequence includes n_orbit, -n_orbit and at least one orthogonal axis. The full experiment uses a one-degree spherical sweep and repeats both drive helicities at every orientation.

11.3 Resonance sweep

A resonant source should produce a response that follows the measured mode excitation rather than the raw generator output. For $SrTiO_3$ this means temperature and frequency dependence around the soft mode. For YIG it means bias-field dependence across ferromagnetic resonance. For CH_3F it means pressure- and frequency-dependent rotational absorption. The response model should be fitted jointly to the internal mode readout and the external force or acceleration readout.

12. Artifact separation and metrology

A microscopic angular-momentum source is driven by strong electromagnetic fields, so the apparatus must be designed around differential symmetry rather than absolute-force readings alone.

Artifact	Parity under helicity reversal	Diagnostic	Mitigation
Absorptive heating	normally even	temperature, pulse energy, time lag	interleaved reversal; matched linear-polarization control
Radiation pressure	normally even for equal incident power	optical/THz power monitor	balanced beams; momentum trap; symmetric mounting
Electrostriction / piezoelectric strain	often even; crystal dependent	strain gauge or interferometer	mechanical symmetry; off-resonance sweep
Lorentz force on conductors	depends on cable currents, not specimen angular momentum	dummy load and cable reversal	remote feed; symmetric current return
Magnetic force in YIG	can be odd under field changes	bias-field-only and microwave-off runs	field-gradient mapping; nonmagnetic control
Acoustic impulse	usually even but phase delayed	microphone or accelerometer array	vacuum enclosure; timing discrimination
Detector electronic pickup	follows drive timing and cables	sensor replacement with electrical dummy	shielding; optical readout; polarity permutation

A valid K-Field source signature must be reconstructed from the full permutation set. It must track the signed microscopic angular momentum, preserve the predicted orientation dependence, follow the material resonance, and remain after subtraction of even-in-helicity electromagnetic backgrounds.

13. Findings and material choices

The highest-priority material is $SrTiO_3$ because its cubic degeneracy supports two orthogonal resonant coordinates in the same mode family, its infrared activity gives direct electric-field coupling, and the resulting ions execute circular real-space trajectories. The drive geometry is exactly the orthogonal-channel architecture: equal in-plane components with relative phase +90 or -90 degrees. The microscopic orbit normal is unambiguous and can be aligned to the K-Field sheet normal.

YIG is the required companion material because it provides a dense coherent source dominated by electronic spin rather than ionic orbit. A paired $SrTiO_3$/YIG program determines whether the response follows total angular momentum, moving nuclear mass or electronic spin.

CH_3F is the best ordinary polar gas in the initial set because its symmetric-top structure, 1.847 D dipole and approximately 51.071 GHz first rotational scale permit direct microwave control with substantially simpler spectroscopy than water. It should follow the solid experiments rather than precede them because the active density and coherence are more sensitive to pressure and cavity conditions.

O_2 and N_2 under an optical centrifuge extend the controlled rotation rate to approximately 10 THz. N_2 serves as the cleaner closed-shell mass-rotation system, while O_2 adds electronic spin. $LiNbO_3$ adds a ferroelectric domain degree of freedom and chiral phonon modes in the 6.8 to 25.4 THz range. Ultracold RbCs establishes the long-coherence limit with 0.78 s Ramsey coherence and greater than 1.4 s spin-echo coherence, but its low particle number makes it a precision reference rather than the first force source.

Recommended starting pair

Primary ionic-orbit specimen: $SrTiO_3$ driven near 3 THz. Primary spin specimen: YIG under bias-tuned microwave resonance. Both use electronically switched quadrature channels and the same helicity-demodulated measurement architecture.

Appendix A. Complete 27-node K-Field Base Cell ledger

Every node is generated from x_L=(1/2)B_nm L. The sheet index is lambda=i+j+k and polarity is chi=(-1)^lambda. Node type is determined by the number of nonzero address components.

L=(i,j,k)	x (nm)	y (nm)	z (nm)	lambda	chi	type
(-1,-1,-1)	-0.263815601790	-0.538717711624	-0.302548870760	-3	-1	corner
(-1,-1,0)	-0.198076662755	-0.291563457733	0.000000000000	-2	+1	edge midpoint
(-1,0,-1)	-0.232828778565	-0.247154253892	-0.302548870760	-2	+1	edge midpoint
(0,-1,-1)	-0.096725762259	-0.538717711624	-0.302548870760	-2	+1	edge midpoint
(-1,-1,1)	-0.132337723721	-0.044409203842	0.302548870760	-1	-1	corner
(-1,0,0)	-0.167089839530	0.000000000000	0.000000000000	-1	-1	face center
(-1,1,-1)	-0.201841955340	0.044409203842	-0.302548870760	-1	-1	corner
(0,-1,0)	-0.030986823225	-0.291563457733	0.000000000000	-1	-1	face center
(0,0,-1)	-0.065738939034	-0.247154253892	-0.302548870760	-1	-1	face center
(1,-1,-1)	0.070364077270	-0.538717711624	-0.302548870760	-1	-1	corner
(-1,0,1)	-0.101350900496	0.247154253892	0.302548870760	0	+1	edge midpoint
(-1,1,0)	-0.136103016305	0.291563457733	0.000000000000	0	+1	edge midpoint
(0,-1,1)	0.034752115809	-0.044409203842	0.302548870760	0	+1	edge midpoint
(0,0,0)	0.000000000000	0.000000000000	0.000000000000	0	+1	body center
(0,1,-1)	-0.034752115809	0.044409203842	-0.302548870760	0	+1	edge midpoint
(1,-1,0)	0.136103016305	-0.291563457733	0.000000000000	0	+1	edge midpoint
(1,0,-1)	0.101350900496	-0.247154253892	-0.302548870760	0	+1	edge midpoint
(-1,1,1)	-0.070364077270	0.538717711624	0.302548870760	1	-1	corner
(0,0,1)	0.065738939034	0.247154253892	0.302548870760	1	-1	face center
(0,1,0)	0.030986823225	0.291563457733	0.000000000000	1	-1	face center
(1,-1,1)	0.201841955340	-0.044409203842	0.302548870760	1	-1	corner
(1,0,0)	0.167089839530	0.000000000000	0.000000000000	1	-1	face center
(1,1,-1)	0.132337723721	0.044409203842	-0.302548870760	1	-1	corner
(0,1,1)	0.096725762259	0.538717711624	0.302548870760	2	+1	edge midpoint
(1,0,1)	0.232828778565	0.247154253892	0.302548870760	2	+1	edge midpoint
(1,1,0)	0.198076662755	0.291563457733	0.000000000000	2	+1	edge midpoint
(1,1,1)	0.263815601790	0.538717711624	0.302548870760	3	-1	corner

Appendix B. Compact control equations

Quantity	Expression
Orthogonal drive	q_x=q0 cos(omega t); q_y=s q0 sin(omega t).
Signed microscopic orbit	L_z=s M_eff omega q0^2.
Coherent angular-momentum density	J_coh=(N/V) M_eff omega q0^2 C_phi C_pop.
K-Field node position	x_L=(1/2) B_nm L, L in {-1,0,1}^3.
K-Field sheet index	lambda=i+j+k.
K-Field sheet plane	g111 dot x=lambda/2, with g111=B_nm^(-T)(1,1,1).
K-Field sheet polarity	chi=(-1)^lambda.
Helicity-odd estimator	S_odd=[S(+)-S(-)]/2.
Helicity-even estimator	S_even=[S(+)+S(-)]/2.
Rigid-rotor first scale for CH3F	f_01 approximately 2 B c.
Photon wavelength	lambda_EM=c/f.

Appendix C. Data provenance and references

Material data are taken from the primary sources below. K-Field geometry is taken from the authoritative Krytronx historical document library. Derived quantities in this report are computed directly from those values using the equations stated in the text and reproduced in the machine-readable companion file.

[1] M. Basini et al., Terahertz electric-field-driven dynamical multiferroicity in SrTiO3, Nature 628, 534-539 (2024). DOI: 10.1038/s41586-024-07175-9. Evidence used: 3 THz circular THz drive, 0.5 THz bandwidth, soft-mode frequencies, field amplitude, ionic displacement estimates and angular-momentum-linked magneto-optical response.

[2] M. Goryachev et al., High-Cooperativity Cavity QED with Magnons at Microwave Frequencies, Physical Review Applied 2, 054002 (2014). DOI: 10.1103/PhysRevApplied.2.054002. Evidence used: YIG sphere, microwave photon-magnon coupling, 2 GHz splitting, cooperativity 10^5 and 3 percent magnetic filling factor.

[3] NIST Computational Chemistry Comparison and Benchmark Database, experimental data for methyl fluoride (CH3F), accessed 2026-07-25. Evidence used: Molecular mass, rotational constants, permanent dipole, polarizability and molecular geometry.

[4] A. Korobenko, A. A. Milner and V. Milner, Direct Observation, Study, and Control of Molecular Superrotors, Physical Review Letters 112, 113004 (2014). DOI: 10.1103/PhysRevLett.112.113004. Evidence used: 792 nm optical centrifuge carrier, rotating polarization accelerated from 0 to 10 THz in about 70 ps, rotational states up to N=95 in O2 and N=60 in N2.

[5] H. Ueda et al., Chiral phonons in polar LiNbO3, Nature Communications 17, 212 (2026). DOI: 10.1038/s41467-025-66911-5. Evidence used: Direct observation of chiral phonons; reversible handedness; representative modes near 28, 76 and 105 meV.

[6] P. D. Gregory et al., Second-scale rotational coherence and dipolar interactions in a gas of ultracold polar molecules, Nature Physics 20, 415-421 (2024). DOI: 10.1038/s41567-023-02328-5. Evidence used: RbCs Ramsey rotational coherence 0.78(4) s; spin-echo lower bound greater than 1.4 s; sample density and dipole data.

[7] Krytronx K-Field historical document library: authoritative K-Field Base Cell matrix and Lucas node/plane conventions. Evidence used: B_nm matrix, 27-node centered address set, reciprocal plane family, sheet spacing and polarity rule.

Document control

Filename: K-Field_materials_for_coherently_controlled_angular_momentum_density_1784992238.pdf
Machine-readable companion: K-Field_materials_for_coherently_controlled_angular_momentum_density_1784992238_machine_readable.json
Unix epoch: 1784992238
Principal Investigator: P. Dwayne Esterline